

HW01-AI

Main Report

Software Testing

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Course: Software Testing

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Date: 2026-05-31

AI Policy: Open – mandatory declaration & AI Audit Report

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1 General Information

Field	Value
Full Name	Tran Hai Duc
Student ID	23127173
Assignment	HW01-AI
Course	Software Testing
Instructors	Dr. Lam Quang Vu; Dr. Tran Duy Hoang; MSc. Tran Thi Bich Hanh; MSc. Truong Phuoc Loc; MSc. Ho Tuan Thanh
Report Date	2026-05-31
AI Policy	Open – mandatory declaration and AI Audit Report attachment

2 Submitted Artifact Summary

Artifact Group	Path
Requirement 1 report	HW1/23127173_HW01_AI_095/requirement/requirement1/requirement1.md
Requirement 1 job data	HW1/23127173_HW01_AI_095/requirement/requirement1/jobs-data/
Requirement 1 AI Mindmap Review	HW1/23127173_HW01_AI_095/requirement/requirement1/AI-Mindmap-Review/ai-mindmap-review.md
Requirement 2 report	HW1/23127173_HW01_AI_095/requirement/requirement2/requirement2.md
Requirement 3 report	HW1/23127173_HW01_AI_095/requirement/requirement3/requirement3.md
Requirement 3 device evidence	HW1/23127173_HW01_AI_095/requirement/requirement3/devices/devices.jpg
Requirement 3 video links	HW1/23127173_HW01_AI_095/requirement/requirement3/video-youtube-test/link-video.md
Requirement 3 AI screenshot	HW1/23127173_HW01_AI_095/requirement/requirement3/screenshot-AI/screenshot-chat-ai.png
Requirement 3 GitHub Issue drafts	HW1/23127173_HW01_AI_095/requirement/requirement3/github-issues/
Requirement 3 GitHub Issues screenshots	HW1/23127173_HW01_AI_095/requirement/requirement3/github-issues/screenshot-defect/
AI Audit Report	HW1/23127173_HW01_AI_095/doc/md/AI Audit/01_AI-Audit-Report.md
AI Critique	HW1/23127173_HW01_AI_095/doc/md/AI Audit/02_AI-Critique.md

Artifact Group	Path
Mandatory Disclosure	HW1/23127173_HW01_AI_095/doc/md/AI Audit/ 03_Mandatory-Disclosure.md
AI Privacy Checklist	HW1/23127173_HW01_AI_095/doc/md/AI Audit/ 04_AI-Privacy-Checklist.md
Prompt Log	HW1/23127173_HW01_AI_095/doc/md/ appendixA-prompt-log.md
Submission Checklist	HW1/23127173_HW01_AI_095/checklist.md

3 Requirement 1 – QA/QC Job Market 2026+

3.1 Scope of Collection

Ten QA/QC job postings were collected within 60 days prior to the collection date of **2026-05-28**. Each posting includes a source link, screenshot with account name, job description, required skills, salary information, and an AI impact analysis. The dataset is compiled from 10 job-*.md files in jobs-data/job-description/ and 10 screenshots in jobs-data/images/.

Collection account displayed in screenshots: Duc Hai - Student at VNUHCM - University of Science.

3.2 Job Posting Summary

ID	Position	Company	Location	Salary	AI / Automation
01	QA – Software / Device Manager	OptiSigns	HCMC	Competitive, undisclosed	Automation testing (bonus)
02	Software QA Dev Engineer / SDET	NVIDIA	Cu Chi, HCMC	Not disclosed	AI tools, model testing, LLM benchmarking
03	QA Engineer	Dwarves Foundation	Hybrid, HCMC	USD 1,800–2,000	ML testing, data-output validation
04	Automation QA Engineer (AI-First)	OPSWAT	HCMC	Not disclosed	AI-first testing, self-healing automation
05	Design Verification Engineer	Hyphen Deux	SHTP, HCMC	Up to USD 5,000	Regression automation, verification tooling

ID	Position	Company	Location	Salary	AI / Automation
06	Full-stack Test Engineer	KMS Technology	Hybrid/Remote	Attractive, undisclosed	AI chat tools, coding assistants, prompt writing
07	Senior QA Tester (Casual Games)	SUN.STUDIO	HCMC	Not disclosed	AI mentioned in hiring support only
08	QC Engineer	Merkle	HCMC	Not disclosed	Prompt regression, AI output evaluation, AI-agent safety
09	Data Test Engineer	HCLTech Vietnam	Hybrid, HCMC	Not disclosed	Automated data validation, data quality tools
10	Quality Assurance Engineer	Quantum Movement	District 3, HCMC	Not disclosed	Automation, computer vision / motion-tracking

3.3 Job Descriptions and Key Skills

ID	Job Description	Key Skills
01	Manage QA for software and devices, build QA strategy, test lifecycle, defect tracking, metrics/reporting, compliance, mentoring.	5+ yr software/hardware QA, lead QA team, test process, automation tools, communication.
02	Create test matrices, test plans, test cases, automation framework, bug lifecycle, verify customer issues/fixes.	3+ yr QA, Linux/Unix, shell/Python, API/UI automation, failure analysis, AI tools, LLM benchmarking.
03	Automation testing API/backend/UI, shift-left, CI/CD, defect tracking, ML output validation, security/auth/API integration.	5+ yr automation, Playwright, Cypress, RestAssured, Postman, Golang/TypeScript/Ruby/Python/Java, Docker.
04	Build automation framework for OESIS, API/system/compatibility testing, CI integration, AI-driven test coverage.	QA automation, Robot Framework/Python, API testing, Git/CI, Agile, AI/ML testing concepts.

ID	Job Description	Key Skills
05	Verification test plan, simulation/emulation/FPGA, debug waveform, coverage analysis, formal verification, regression.	Verilog/SystemVerilog, UVM, Perl/Python/TCL, PCIe/USB/DDR, CDNS EDA.
06	Manual/automation testing, automate functional/regression/performance tests, increase end-to-end coverage.	API/web/mobile testing, Selenium/Playwright, C#/Java/JS/Python, Postman, Git, AI tools (nice-to-have).
07	Ensure no critical bugs at release, test ads/analytics/localization/ANR/UX/performance/device compatibility.	Senior game testing, test checklists/templates, bug lifecycle, Ucommunication, ISTQB preferred.
08	Establish QC process for Salesforce CRM and AI solutions, automation framework, AI testing procedure, quality reporting.	2–3 yr QC/QA, Selenium/Playwright/Cypress, API testing, Salesforce, AI tools, prompt regression.
09	Data testing for banking project, ETL testing, AWS data services, data warehouse and data quality framework.	Python/Rust/Go/Java, SQL, AWS Lambda/Glue/Athena/S3, Redshift, dbt, Great Expectations, Soda.
10	Test mobile/web app, hardware integration, sensor responsiveness, video recording stability, backend API integration.	7+ yr QA, Selenium/Appium/XCUITest, REST API, Flutter, ReactJS, Flipper, Android Studio Profiler.

3.4 AI Impact Analysis

ID	AI Impact Analysis
01	AI can aggregate defect trends, QA metrics, and release risks, but the QA manager still makes strategy and mentoring decisions.
02	AI expands QA into model testing and LLM benchmarking, requiring testers to understand both automation and model behaviour.
03	ML/data validation requires clear oracles; AI can generate tests, but QA must determine correctness, safety, and stability of outputs.
04	AI-first testing increases coverage and reduces automation maintenance, but generated/self-healing tests must be reviewed.
05	AI can assist with log reading and coverage-gap suggestions, but design verification requires technical evidence and domain knowledge.

ID	AI Impact Analysis
06	AI tools support research, debugging, test snippets, and documentation; QA must still validate output and protect sensitive data.
07	AI can generate checklist ideas, but game QA requires real playtesting, UX perception, and device/performance validation.
08	QC for AI solutions must include prompt regression, output quality, model degradation, and AI-agent safety before go-live.
09	AI can generate SQL/data checks, but banking data testing requires traceability and reproducible results.
10	AI can suggest mobile automation and profiler log analysis, but senior QA must verify on real devices, sensors, and performance.

3.5 Requirement Compliance

Criterion	Status
10/10 postings are QA/QC/Tester/Verification/Data Test roles	Achieved
Each posting has a source link and screenshot in the repository	Achieved
Each posting has job description and required skills in job-*.md	Achieved
≥3 postings mention AI/LLM/Automation-AI explicitly	Achieved (Jobs 02, 03, 04, 06, 08)
Salary disclosed per JD (Job 03: USD 1,800–2,000; Job 05: up to USD 5,000; others: undisclosed)	Partial

Key Observation: The 2026+ QA/QC market extends well beyond traditional manual and automation testing to encompass SDET, AI/LLM testing, data testing, Salesforce/AI solution QC, device/mobile QA, and design verification. AI assists with test generation, log analysis, documentation, and metric aggregation; however, QA engineers remain responsible for verifying results, protecting data, and owning release decisions.

4 Requirement 2 – 20 Software Defects (2022–2026)

4.1 Scope

Twenty publicly documented software defects from 2022–2026 were compiled. Each entry includes a source link, description, severity, consequences, remediation, and a note on how AI might produce biased or hallucinated explanations. The list prioritises defects with public sources across security vulnerabilities, outages, cloud misconfiguration, AI hallucination, prompt injection, AI bias, and data exposure.

4.2 Defect Summary Table

ID	Year	Defect / Incident	Severity	Consequences	Remediation	AI Bias / Hallucination Risk
01	2024	CrowdStrike Falcon Channel File 291 outage	Critical	Millions of Windows BSOD; aviation, healthcare, banking disrupted.	Staged rollout, canary, content validation, auto-rollback.	AI may hallucinate this as a Microsoft cyberattack; it was a content-update defect in CrowdStrike.
02	2023	FAA NOTAM outage	Critical	Thousands of US flights delayed/grounded	Change control, backup/restore rehearsal, DB sync tests.	AI may bias toward “aging system” and miss the specific change-management failure.
03	2022	Atlassian Confluence CVE-2022-26134 (OGNL RCE)	Critical	Servers compromised, webshells installed, crypto-mining, data theft.	Upgrade, WAF rules, IOC scanning, rotate secrets.	AI may hallucinate this as XSS because both involve input injection; must clarify OGNL/RCE.
04	2022	OpenSSL CVE-2022-3602 / CVE-2022-3786 (buffer overflow)	High	DoS or potential RCE depending on environment.	Update OpenSSL, rebuild containers/base images.	AI may hallucinate this as “Heartbleed 2”; different CVE, different mechanism.
05	2023	MOVEit Transfer CVE-2023-34362 (SQL injection)	Critical	Data exfiltration across many organisations, ransomware extortion.	Emergency patch, disable public access, forensic scan, rotate credentials.	AI may hallucinate the issue as SFTP encryption; root cause is SQL injection in the web app.

ID	Year	Defect / Incident	Severity	Consequences	Remediation	AI Bias / Hallucination Risk
06	2024	XZ Utils backdoor CVE-2024-3094	Critical	Risk of remote unauthorised access on affected Linux systems.	Roll back to safe version, pin dependencies, verify tarballs, harden maintainer review.	AI may bias toward “open source is insecure”; the lesson is supply-chain governance.
07	2023	Toyota cloud misconfiguration data exposure	High	Customer/connected car data exposed; privacy and reputation impact.	Cloud posture management, least privilege, public-bucket detection, periodic audit.	AI may hallucinate a deliberate attack; public sources confirm mis-configuration.
08	2023	ChatGPT Redis bug exposed chat/payment data	High	Chat titles and payment metadata of some ChatGPT Plus users exposed.	Patch library, test race conditions, cache isolation, postmortem, user notification.	AI may hallucinate the model “self-disclosed” training data; the real bug was a system/cache issue.
09	2024	Google Gemini image generation bias	Medium	Loss of trust, image generation feature suspended.	Context-aware safety rules, bias tests, human review, diverse evaluation datasets.	AI may bias that “diversity is a bug”; the real issue is applying rules out of context.
10	2024	Air Canada chatbot hallucinated refund policy	High	Air Canada held legally liable for chatbot mis-information.	RAG with accurate policy sources, response citation, guardrails, human escalation.	This is a direct hallucination: AI generated wrong policy and presented it as authoritative.

ID	Year	Defect / Incident	Severity	Consequences	Remediation	AI Bias / Hallucination Risk
11	2024	DPD chatbot jailbreak / prompt injection	Medium	Brand damage, DPD partially disabled the chatbot.	Prompt-injection hardening, output moderation, role constraints, safe fallback.	AI may hallucinate the chatbot “spontaneously” misbehaved; it was prompted/jailbroken after a faulty update.
12	2023	Chevrolet dealer chatbot agreed to \$1 Tahoe	Medium	Viral incident, brand damage, chatbot removed.	Prevent LLM from quoting/contracting rule-based boundaries, legal disclaimers.	AI may hallucinate this as a legally binding contract; the bot had no legal authority to sell a car.
13	2023	Samsung employees leaked confidential data to ChatGPT	High	IP and confidential data exposed; Samsung tightened AI policy.	Enterprise AI policy, DLP, prompt logging, private/on-prem model, employee training.	AI may blame ChatGPT entirely; the incident is a shadow-AI/process control and user data-handling failure.
14	2023	Microsoft AI researchers exposed 38 TB via SAS token	High	38 TB of internal private data potentially accessible/modifiable/	Least-privilege SAS, short-lived tokens, secret scanning, CI policy, access review.	AI may hallucinate a model leak; reality is cloud access-token misconfiguration by the AI team.
15	2024	Microsoft Recall privacy/security design issue	High	Microsoft delayed rollout; heavy criticism on privacy/security.	Opt-in, encryption, local auth, sensitive-data filtering, red-team, privacy threat modelling.	AI may bias that “on-device is safe”; on-device still has risk if malware or another user accesses the data.

ID	Year	Defect / Incident	Severity	Consequences	Remediation	AI Bias / Hallucination Risk
16	2024	Polyfill.io supply-chain attack	Critical	Hundreds of thousands of websites at risk of redirect/malware.	Remove polyfill.io, self-host, SRI/CSP, dependency inventory, third-party script monitoring.	AI may hallucinate an npm package bug; reality is a third-party CDN/domain supply-chain takeover.
17	2025	DeepSeek exposed database with chat history/API data	High	Prompts, tokens/API keys, internal details, and user privacy at risk.	Authenticate the database, network isolation, secret rotation, logging hygiene.	AI may introduce political bias about “Chinese AI”; analysis should focus on the misconfigured database.
18	2023	23andMe credential stuffing/data exposure	High	Personal and genetic data of millions of users affected.	Mandatory MFA, password reset, rate limiting, bot detection, anomaly detection.	AI may hallucinate a SQL injection; 23andMe stated no evidence of a breach in their own systems.
19	2023	CitrixBleed CVE-2023-4966	Critical	Session hijacking, LockBit ransomware activity, unauthorised network access.	Patch, terminate active sessions, rotate credentials/tokens, IOC hunting, segmentation.	AI may say patching is sufficient; CitrixBleed requires killing sessions and rotating tokens because tokens were already stolen.
20	2024	OpenSSH regreSSHion CVE-2024-6387	Critical	Remote attacker may gain root access on affected systems.	Upgrade OpenSSH, reduce LoginGraceTime, restrict network exposure, scan assets.	AI may hallucinate this is a completely new SSH bug; it is a regression of a bug patched in 2006.

4.3 Statistics

Metric	Result
Total defects	20
Period covered	2022–2026
AI/LLM-related defects	9/20 (IDs 08, 09, 10, 11, 12, 13, 14, 15, 17)
AI/LLM categories	Hallucination, prompt injection/jailbreak, bias, data exposure, AI privacy/security design
All 20 entries have AI bias/hallucination note	Achieved

Key Observation: Defects from 2022–2026 show that modern QA/QC must address not only functionality but also security, data integrity, deployment processes, supply-chain integrity, and AI behaviour. For AI/LLM systems, bugs often reside in prompts, retrieval data, guardrails, tool access permissions, privacy policies, and model reasoning paths. QA must combine traditional test cases with threat modelling, data validation, red-team prompt testing, and post-deployment monitoring.

5 Requirement 3 – Test Cases for a Physical Product

5.1 Device Under Test

The selected product is the **Casper Remote U25 Series** air-conditioner remote control. The remote features an LCD display and the following main buttons: Power, Mode, Speed, Turbo, iSAVE, Baby Care, Temperature +/-, L/R Swing, U/D Swing, Menu, and OK.

5.2 Test Assumptions

- The remote has batteries installed and the LCD display is functional.
- The corresponding air conditioner is operating normally and receiving IR signals from the remote.
- The tester stands at close range, pointing the remote at the AC receiver, unless the test case specifies blocking the signal or changing distance/angle.

Field	Value
Product	Air-conditioner remote control
Brand	Casper
Model	Remote U25 Series
Year of manufacture	Not visible on device evidence
Serial number	Not visible on device evidence
Device photo	devices/devices.jpg – remote + student ID card in same frame
Video evidence	video-youtube-test/link-video.md – 5 YouTube Shorts links

- Actual results and defect conclusions are based solely on real-device testing. The table below contains 15 initial test cases and 3 supplementary edge cases discovered by the student.

5.3 Test Case Suite

TC ID	Objective	Input	Steps	Expected Result	Actual Result	Verdict
TC-01	Verify basic power on/off via Power button.	Power button (red).	1. Point remote at AC. 2. Press Power to turn on. 3. Wait for AC response. 4. Press Power again to turn off.	AC turns on/off accordingly; remote shows clear power state; AC beeps or responds.	As expected.	Pass
TC-02	Verify switching to Cool mode.	Mode button → Cool.	1. Turn on AC. 2. Press Mode until LCD shows C00L. 3. Wait 10–20 s.	LCD shows Cool; AC switches to cooling; fan/flap responds stably.	As expected.	Pass
TC-03	Verify temperature increment/decrement within valid range.	+ and - buttons.	1. Enable Cool. 2. Press + repeatedly to upper limit. 3. Press - repeatedly to lower limit.	Temperature changes step by step; does not exceed AC's supported range; LCD does not freeze or show garbage.	As expected.	Pass
TC-04	Verify Speed button cycles fan speed.	Speed button.	1. Enable Cool. 2. Press Speed repeatedly. 3. Observe fan-speed icon and actual fan.	Each press cycles to next speed in loop; AC responds correctly.	Fan audibly quieter at silent, louder at turbo.	Pass

TC ID	Objective	Input	Steps	Expected Result	Actual Result	Verdict
TC-05	Verify Turbo mode.	Turbo button.	1. Enable Cool. 2. Press Turbo. 3. Wait 10–20 s. 4. Press Turbo again to disable if applicable.	Turbo engages/disengages clearly; AC increases capacity/fan as designed; LCD shows corresponding icon.	As expected.	Pass
TC-06	Verify L/R Swing.	L/R Swing button.	1. Turn on AC. 2. Press L/R Swing. 3. Observe left/right louver movement. 4. Press again to stop/reposition.	Left/right louver moves or changes state as commanded; no sticking.	As expected.	Pass
TC-07	Verify U/D Swing.	U/D Swing button.	1. Turn on AC. 2. Press U/D Swing. 3. Observe up/down louver movement. 4. Press again to stop/reposition.	Up/down louver moves or changes state as commanded; no sticking.	As expected.	Pass
TC-08	Verify iSAVE saves/restores configuration.	Temperature Mode, Speed, iSAVE button.	1. Set Cool, temperature X, speed Y. 2. Press iSAVE. 3. Change to different config. 4. Press iSAVE again.	Remote/AC restores saved config or activates energy-saving mode per manual.	iSAVE only saves initial state; re-pressing iSAVE does not update saved config.	Fail
TC-09	Verify Baby Care does not conflict with Turbo.	Baby Care and Turbo buttons.	1. Enable Cool. 2. Press Baby Care. 3. Press Turbo. 4. Observe LCD and AC response.	System handles priority clearly: either prevents both from being active simultaneously or disables conflicting mode; no contradictory LCD state.	As expected.	Pass

TC ID	Objective	Input	Steps	Expected Result	Actual Result	Verdict
TC-10	Verify Menu/OK navigates sub-features.	Menu and OK buttons.	1. Press Menu. 2. Navigate with directional keys if available. 3. Press OK. 4. Wait for timeout.	Menu displays/changes items correctly; OK confirms action; idle timeout exits menu safely.	As expected.	Pass
TC-11	Verify rapid repeated Power presses.	Power button $\times 5$ in 3 s.	1. Point remote at AC. 2. Press Power 5 times rapidly. 3. Observe LCD and AC.	Final state matches press count; remote does not freeze; AC not left in half-on/half-off state.	As expected.	Pass
TC-12	Verify holding +/- beyond limits.	Hold + 5 s, hold - 5 s.	1. Enable Cool. 2. Hold + to upper limit, continue 5 s. 3. Repeat with -.	Remote clamps at limit; no overflow; no unexpected value jump.	As expected.	Pass
TC-13	Verify commands are rejected when IR emitter is blocked.	Hand/paper blocking IR emitter.	1. Turn on AC. 2. Block IR emitter on remote. 3. Press Mode or Speed. 4. Unblock and press again.	When blocked, AC does not change state (LCD may still update); after unblocking, new commands are received normally.	AC still responds normally even with emitter blocked.	Fail
TC-14	Verify limited angle/distance acceptance.	Distance 5–7 m, off-axis angle.	1. Stand 5–7 m from AC. 2. Tilt remote left/right. 3. Press Power/Mode. 4. Repeat pointing straight ahead.	Commands only reliable within supported angle/distance; out-of-range attempts fail clearly, not silently.	AC responded at all tested angles without failure.	Fail
TC-15	Verify LCD dim and low-battery indicator.	Weak/near-empty batteries.	1. Observe LCD with current batteries. 2. If low battery present, try sending Power/Mode. 3. Replace with fresh batteries and repeat.	Remote shows clear low-battery indication; commands do not intermittently misfire; fresh batteries restore stability.	Batteries still full; test deferred.	Pending

TC ID	Objective	Input	Steps	Expected Result	Actual Result	Verdict
TC-16	<i>Edge case:</i> AC should only power on via Power button.	AC off; press Turbo, Mode, Speed.	1. Confirm AC is off. 2. Do not press Power. 3. Press Turbo, then Mode, then Speed. 4. Observe AC and remote LCD.	AC must not power on when only feature buttons are pressed; only Power button turns on the AC.	Pressing Turbo/Mode turned the AC on.	Fail
TC-17	<i>Edge case:</i> Switching from Cool to Dry automatically adjusts fan speed.	In Cool mode; press Mode to switch to Dry.	1. Turn on AC in Cool mode. 2. Set fan speed above minimum. 3. Press Mode to switch to Dry. 4. Observe fan speed after mode change.	When switching to Dry, AC automatically adjusts fan speed to a level appropriate for dehumidification.	Fan automatically dropped to minimum speed.	Pass
TC-18	<i>Edge case:</i> Baby Care locks fixed configuration.	Baby Care active; press +, -, L/R Swing, U/D Swing.	1. Turn on AC. 2. Press Baby Care. 3. Try to increase/decrease temperature. 4. Try to change airflow direction. 5. Observe LCD and AC.	While Baby Care is active, protective/comfort settings are fixed; temperature and airflow direction changes are not permitted.	Could not increase/decrease temperature or change airflow direction while Baby Care was active.	Pass

5.4 Videos with Evidence

Video	TC	Evidence / Link
V1	TC-01	YouTube Shorts – demonstrates real Power command sent by remote.
V2	TC-02	YouTube Shorts – demonstrates Cool mode on LCD and AC.
V3	TC-03	YouTube Shorts – demonstrates temperature increment/decrement within limits.
V4	TC-05	YouTube Shorts – demonstrates Turbo button.
V5	TC-18	YouTube Shorts – demonstrates Baby Care locking configuration.

5.5 Defect Log

The defects below were derived from test cases with a **Fail** verdict: TC-08, TC-13, TC-14, and TC-16. Test cases with Pass or Pending verdicts are not counted as defects to avoid recording unsupported issues.

Defect	TC	Summary	Expected	Actual	Severit	Status
D-01	TC-08	iSAVE does not save/update most recent configuration.	iSAVE saves or restores config as user intends.	iSAVE only stores the first state; subsequent presses do not update saved config.	Medium	Open
D-02	TC-13	Remote operates normally even when IR emitter is physically blocked.	AC should not receive commands when IR emitter is blocked.	AC responds normally despite emitter being blocked.	Low	Open
D-03	TC-14	AC accepts commands at all tested angles without showing any range limitation.	Commands should fail or be unreliable outside supported angle/distance.	Commands accepted at all tested angles.	Low	Open
D-04	TC-16	Feature buttons (Turbo/Mode/i can power on the AC while it is off.	Only the Power button should turn on the AC.	Pressing Turbo, Mode, or Speed turns on the AC.	High	Open

Note: The requirement target is ≥ 5 confirmed defects; currently only 4 have been confirmed. No defects were fabricated. Additional testing must be conducted, and defects may only be added upon real-world evidence.

5.6 Student-Discovered Edge Cases (AI-Missed)

A screenshot of the AI chat session is provided in `screenshot-AI/screenshot-chat-ai.png` to confirm that these edge cases were not generated in the AI baseline prompt. This screenshot is genuine evidence collected manually, not generated by AI.

TC	Edge Case Description	Why AI Tends to Miss It
TC-16	AC should only power on via the Power button; however, feature buttons (Turbo/Mode/Speed) can also turn the unit on.	AI typically treats Turbo/Mode/Speed as post-power-on feature buttons, so it overlooks the precondition “AC is off” and the unintended power-on behaviour.
TC-17	Switching from Cool to Dry automatically reduces fan speed to minimum.	AI usually only checks that the LCD changes mode and does not test the side-effect of mode change on fan speed.
TC-18	Baby Care mode locks temperature and airflow direction controls.	AI typically only tests Baby Care on/off toggling; it misses the safety/comfort constraint that prevents user adjustments while Baby Care is active.

5.7 GitHub Issues for Defect Screenshots

Defects are logged as GitHub Issues in the personal repository `HappyDuckCoder/Software-Testing`. Draft files were created locally and then posted manually.

Defect	Local Draft File	Status
D-01	<code>github-issues/D-01-isave-not-saving-latest-config.md</code>	GitHub Issue #1 created; screenshot: <code>D01.png</code>
D-02	<code>github-issues/D-02-remote-works-when-ir-blocked.md</code>	GitHub Issue #2 created; screenshot: <code>D02.png</code>
D-03	<code>github-issues/D-03-commands-work-at-all-tested-angles.md</code>	GitHub Issue #3 created; screenshot: <code>D03.png</code>
D-04	<code>github-issues/D-04-feature-buttons-power-on-ac.md</code>	GitHub Issue #4 created; screenshot: <code>D04.png</code>

Issue links are recorded in `github-issues/github-issues-links.md`. The issues list screenshot at `screenshot-defect/list-defect.png` displays the GitHub username `HappyDuckCoder`.

5.8 Requirement Compliance

6 AI CLO / Bloom-AI Outcomes

Criterion	Status
One specific household device selected	Achieved
Device photo + student ID card	Achieved
Brand/model; year/serial (partially visible)	Partial
15+ test cases with Objective/Input/Steps/Expected/Actual/Verdict	Achieved (18 total)
≥ 5 test cases with video ≤ 60 s	Achieved (5 YouTube Shorts: TC-01,02,03,05,18)
≥ 3 edge cases AI did not generate	Achieved (TC-16, TC-17, TC-18)
≥ 5 confirmed defects from device	Not Achieved (4 confirmed: D-01 to D-04)
AI not used to generate prohibited evidence	Achieved

CLO	Requirement	Evidence	Status
G9.1 – Understand	Request AI to generate a QA/QC role mindmap, then identify errors.	AI-Mindmap-Review/ ai-mindmap-review.md	Achieved
G9.3 – Analyse	Analyse AI output and find ≥ 3 edge cases that AI missed.	TC-16, TC-17, TC-18; screenshot-AI/ screenshot-chat-ai.png	Achieved

In the mindmap review, the initial AI output omitted: Data Test Engineer, Design Verification, Device/Mobile QA, and the distinction between AI-assisted QA and AI/LLM system testing. The student confirmed these omissions as valid deficiencies after cross-referencing all 10 jobs from Requirement 1 and produced a corrected mindmap.

7 AI Audit Report – Full Content

7.1 Student and AI-use Information

7.2 Audit Table

Field	Value
Student name	Tran Hai Duc
Student ID	23127173
Assignment	HW01-AI
Updated date	2026-05-31
AI tools used	Codex / ChatGPT; Claude; Pandoc for TeX conversion
AI usage policy	Open, mandatory disclosure and audit required

#	Prompt / Tool	AI Output	Verdict	Reason and Student Verification
1	Codex / ChatGPT, 2026-05-28: read AI templates, job links, screenshots, update CSV, Requirement 1, AI Audit, policies.	Initial drafts for job Markdown files, job summary CSV, Requirement 1, and AI Audit.	INCOMPLETE	Needed comparison with real screenshots and job descriptions. Student later corrected links, screenshots, job details, and salary notes.
2	Codex / ChatGPT with web crawl, 2026-05-28: crawl/search job details and update Requirement 1.	Public job detail crawl/search and dataset/report update.	INCOMPLETE	Public pages may differ from screenshots or LinkedIn view. Student manually reviewed and edited the 10 job files.
3	Codex / ChatGPT, 2026-05-28: reformat the 10 job-id files, summary, Requirement 1, and audit.	Reformatted 10 job Markdown files, regenerated CSV, shortened Requirement 1, updated audit/disclosure/prompt log.	INCOMPLETE	AI helped structure JD data, but student must verify screenshots, posting date, salary, and link fields.
4	Codex / ChatGPT with web search, 2026-05-28: do Requirement 2 and audit.	20 software defects from 2022–2026 with source, description, severity, consequence, fix, and AI bias/hallucination note.	INCOMPLETE	Incidents/advisories must be checked against primary sources to avoid wrong year, severity, or cause.
5	Codex / ChatGPT, 2026-05-31: do Requirement 3 for the air-conditioner remote and check policy.	Test-design draft for Casper remote: device declaration, 15 test cases, video suggestions, defect candidates, and AI-missed edge cases.	INCOMPLETE	AI cannot replace testing on the physical device and cannot create prohibited evidence such as photos, videos, or real results.

#	Prompt / Tool	AI Output	Verdict	Reason and Student Verification
6	Codex / ChatGPT, 2026-05-31: add device information: Casper, Remote U25 Series, device photo.	Updated Requirement 3 and device info with brand/model/photo; marked year and serial as not visible.	INCOMPLETE	AI only inserted information provided by the student; no serial/year/actual result/video was fabricated.
7	Codex / ChatGPT, 2026-05-31: add TC16–TC18 student-found edge cases.	Added TC16–TC18 as official edge cases, kept TC11–TC15 as normal test cases, updated actual/verdict.	INCOMPLETE	Actual results for TC16–TC18 came from student observation; AI only formatted them.
8	Codex / ChatGPT, 2026-05-31: update Requirement 3 with YouTube links.	Inserted 5 YouTube Shorts links for TC-01, TC-02, TC-03, TC-05, and TC-18.	INCOMPLETE	AI inserted links supplied by the student but did not verify duration, voice, or accessibility of each video.
9	Codex / ChatGPT, 2026-05-31: rewrite defects from failed cases.	Rewrote defect log as 4 confirmed defects from TC-08, TC-13, TC-14, and TC-16.	INCOMPLETE	AI converted failed test cases into structured defect reports; it did not invent a fifth defect.
10	Codex / ChatGPT, 2026-05-31: review Requirement 1 and create checklist.	Created checklist for Requirement 1: files, screenshots, CSV, report, salary, AI impact, and final verification items.	INCOMPLETE	AI can check text/files but cannot visually guarantee every screenshot contains account name/date unless manually reviewed.
11	Codex / ChatGPT, 2026-05-31: extend checklist for Requirement 2 and 3.	Expanded checklist with Requirement 2 and Requirement 3 compliance, evidence, and unresolved risks.	INCOMPLETE	Student still must manually verify videos, YouTube access, screenshots, and GitHub Issues.
12	Codex / ChatGPT, 2026-05-31: add AI Mindmap Review.	Created AI mindmap review with baseline AI output, 4 AI omissions/errors confirmed by student, corrected mindmap, and G9.1/G9.3 mapping.	INCOMPLETE	Student must ensure the critique reflects their own comparison with the 10 job postings.

#	Prompt / Tool	AI Output	Verdict	Reason and Student Verification
13	Codex / ChatGPT, 2026-05-31: create final main report from Requirements 1–3 and audit files.	Drafted main report with student information, instructors, R1/R2/R3 summary, AI CLO, audit summary, critique, disclosure, self-assessment, and signature.	INCOMPLETE	Main report is a synthesis; evidence and links still require final student verification.
14	Codex / ChatGPT, 2026-05-31: rewrite Requirement 1/2/3 with Vietnamese accents.	Rewrote the three requirement Markdown files with Vietnamese accents while preserving IDs, links, and technical terms.	INCOMPLETE	Student must review that accent conversion did not change technical meaning, verdicts, severity, salary, or sources.
15	Claude, 2026-05-31: generate English LaTeX from Markdown report.	Produced/proposed an English LaTeX report draft from the Markdown main report.	INCOMPLETE	Translation/conversion may lose paths, alter meaning, or desynchronise audit and defect counts.
16	Codex / ChatGPT, 2026-05-31: create local GitHub Issue drafts for confirmed defects.	Created four local issue drafts and <code>github-issues-link</code> updated reports/audit to note real GitHub Issues were still required.	INCOMPLETE	Local drafts are not real GitHub Issues and cannot replace username screenshots.
17	Codex / ChatGPT, 2026-05-31: update after real GitHub Issue screenshots were added.	Updated links, defect drafts, Requirement 3, checklist, main report, and audit for issues #1–#4 in <code>HappyDuckCoder/Software-Testing</code> .	INCOMPLETE	AI only read file names/observed screenshots; student must keep original screenshots and ensure repo access remains valid.
18	Codex / ChatGPT, 2026-05-31: embed full Requirement 1/2/3 content into main report.	Added full R1, R2, R3 content to the Markdown main report: job tables, AI impact, 20 defects, 18 test cases, videos, defects, edge cases, policy, GitHub Issues.	INCOMPLETE	AI only merged existing repository content; student must check paths, tables, Vietnamese text, and PDF readability.

#	Prompt / Tool	AI Output	Verdict	Reason and Student Verification
19	Codex / ChatGPT + Pandoc, 2026-05-31: synchronise TeX/PDF from Markdown after full R1/R2/R3 embedding.	Regenerated TeX and built PDF from the Markdown main report.	INCOMPLETE	Pandoc converts format only; student must inspect layout, page breaks, links, fonts, and wide tables.
20	Codex / ChatGPT + Pandoc, 2026-05-31: add full AI Audit to main report, not only summary; update Markdown and LaTeX without PDF build.	Embedded full AI Audit Report, AI Critique, Mandatory Disclosure, and Privacy Checklist into main report; synchronised LaTeX content without rebuilding PDF.	INCOMPLETE	AI only merged existing audit content and TeX text; student must review headings, long tables, and disclosure visibility.

7.3 AI Accuracy Summary

Metric	Result
Total AI-generated artifacts audited	20
VALID	0
INVALID	0
INCOMPLETE	20

All AI-supported artifacts are marked **INCOMPLETE** because AI was used for drafting, formatting, translation, conversion, aggregation, and suggestions only. The student remains responsible for verifying primary evidence: job screenshots, source links, videos, device photo, actual results, GitHub Issues, username screenshots, and defect evidence.

7.4 When AI Should and Should Not Be Used

AI is suitable for standardising format, summarising long job descriptions, detecting missing fields, suggesting test ideas, and turning rough observations into structured tables. AI must not be used to create or replace original evidence such as screenshots, account names, posting dates, salary evidence, source links, device photos, test videos, actual results, GitHub Issues screenshots, or defect proof. For HW01, AI is an editorial/checking assistant; the student is responsible for source verification, real-device testing, and final submission integrity.

8 AI Critique – Full Content

In HW01, AI was most useful in an editorial and structural role: standardising 10 JDs from Requirement 1 into Markdown/CSV, aggregating the 20 software defects for Requirement 2, proposing test cases for the air-conditioner remote in Requirement 3, scaffolding the G9.1 mindmap, converting defect logs into GitHub Issue drafts, updating the report after the student added real GitHub Issue screenshots, embedding full Requirement 1/2/3 content into the main report, synchronising the L^AT_EX/PDF from Markdown, and embedding full AI Audit content into the main report instead of leaving only a summary. Nevertheless, the audit shows all 20 artifacts should be considered **INCOMPLETE** until primary evidence is verified by the student. AI can write coherently, but it cannot view screenshots, guarantee link validity, run tests on real hardware, or substitute for prohibited evidence such as device photos, videos, actual results, real GitHub Issues, or defect proof.

AI errors fell into three main categories:

1. **Over-inference:** When job descriptions were ambiguous, AI inferred AI skills for generic automation roles or interpreted undisclosed salaries without sufficient basis.
2. **Representation bias:** AI training skews toward web/software QA patterns, so the initial mindmap omitted data testing, design verification, device/mobile QA, and AI-agent safety. The student confirmed these omissions as valid after cross-referencing the 10 collected jobs.
3. **Untested physical states:** For the air-conditioner remote, AI generated well-structured button-press test cases but overlooked physical preconditions such as the unit being powered off (TC-16), mode-transition side effects on fan speed (TC-17), and configuration-lock behaviour under Baby Care (TC-18).

Lesson Learned: AI must not be treated as the final source of truth. The correct workflow is: *let AI draft* → *separate AI output from primary evidence* → *cross-reference against screenshots/links/real hardware* → *record audit verdicts* → *conclude Pass/Fail/defect only upon real-world observation*. AI accelerates the process; quality assurance and academic integrity remain the student's responsibility.

9 Mandatory Disclosure – Full Content

Requirement 1 dataset, Requirement 2 software-defect dataset, Requirement 3 test-design draft, summary tables, CSV, mindmap review, checklist, main report, L^AT_EX translation draft, GitHub Issue draft text, and draft report sections were generated and formatted with assistance from Codex / ChatGPT and Claude. I provided or corrected source job details for Requirement 1, reviewed the generated Requirement 2 defect list, sources, severity, consequences, fixes, and AI bias/hallucination notes, and reviewed/modified the Requirement 3 remote air-conditioner test cases.

I added student-found edge cases TC-16, TC-17, and TC-18 about off-state feature buttons, Cool-to-Dry fan adjustment, and Baby Care fixed configuration. The LinkedIn screenshots, source links, physical device photo, videos, actual results, real GitHub Issues, GitHub Issues screenshot, and defect evidence are collected or verified manually by me. The detailed AI Audit Report is attached in `doc/md/AI Audit/01_AI-Audit-Report.md`. I confirm I did not use AI to generate any prohibited artifact, including job posting screenshots with account name, physical device photos, videos, GitHub Issues screenshots, or fake evidence.

Field	Value
Full name	Tran Hai Duc
Student ID	23127173
Assignment	HW01-AI
Date	2026-05-31
Signature	Tran Hai Duc

10 AI Privacy and Responsible Use Checklist – Full Content

10.1 Before Using AI

- ✓ Confirmed the assignment permits AI use under an Open policy with mandatory disclosure.
- ✓ Read the AI Audit Report requirement in `HW1/requirement/HW1-requirement.md`.
- ✓ Understood prohibited AI-generated artifacts: job screenshots with account name, device photo with student ID, test videos, GitHub Issues screenshots with username, and prompt-log timestamps.

10.2 While Using AI

- ✓ Did not input sensitive third-party personal data.
- ✓ Did not ask AI to create fake screenshots, fake links, or fake evidence.
- ✓ GitHub Issue drafts assisted by AI are only draft text; real issues and username screenshots were created/verified manually.
- ✓ Recorded main prompts and outputs in the AI Audit Report.
- ✓ Marked AI-inferred parts when screenshots did not contain enough detail.
- ✓ Did not ask AI to create the remote + student ID photo, videos, actual results, or defect evidence for Requirement 3.

10.3 Before Submission

- ✓ Every AI-supported artifact has an audit row.
- ✓ Unverified fields are marked as **Not disclosed**, **Not visible**, or data-quality notes.
- ✓ The 10 job-id files are standardised from QA/QC/Tester/Verification/Data Test job descriptions.
- ✓ Requirement 3 includes the real remote + student ID photo in `requirement3/devices/devices.jpg`.
- ✓ Requirement 3 includes 5 video links in `requirement3/video-youtube-test/link-video.md`.
- ✓ Requirement 3 includes real GitHub Issues and screenshots showing the GitHub username.
- Student should still verify video access, source links, screenshot account names, and final PDF layout before submission.
- Student should fill class/faculty information if the official submission form requires it.

10.4 Final Responsibility Statement

Final responsibility for accuracy, originality, evidence authenticity, and academic integrity belongs to the student. Any undeclared AI use is considered an academic integrity violation.

11 Self-Assessment

#	Item	Max	Self	Notes
1	Job Market 2026+	40	38	10 jobs, screenshots, sources, CSV, AI impact; salary and screenshot items need one final check.
2	Software Defects 2022–2026	20	20	20 public defects with sources, severity, consequences, remediation, and AI bias/hallucination notes.
3	Physical-product test design	25	22	Device, photo, 18 test cases, 5 videos, GitHub Issues with username screenshot; primary risk: only 4 confirmed defects vs. ≥ 5 target.

#	Item	Max	Self	Notes
AI-1	AI Audit Report	8	8	AI Audit Report present; 20 AI-assisted artifacts logged.
AI-2	AI Critique + Disclosure	4	4	Critique, mandatory disclosure, and AI declaration complete.
AI-3	AI Checklist + Anti-cheat artifacts	3	3	Checklist, privacy checklist, prompt log, real photo/video/screenshot evidence all present.
Total		100	95	Proposed self-score: 95 / 100

12 Declaration and Signature

I hereby declare that all prohibited evidence — including job posting screenshots with account name, device photos with student ID, test videos, actual results, and defect evidence — was **not** generated by AI. All AI-assisted sections have been declared in the AI Audit Report and Appendix A Prompt Log.

Field	Value
Full Name	Tran Hai Duc
Student ID	23127173
Date Signed	2026-05-31
Signature	<i>Tran Hai Duc</i>